

Semester: Fall 2015
Linear Circuit Analysis

Quiz-2(B) section EE-2

Student Roll Number: 15L-4522

Time allowed: 15 minutes

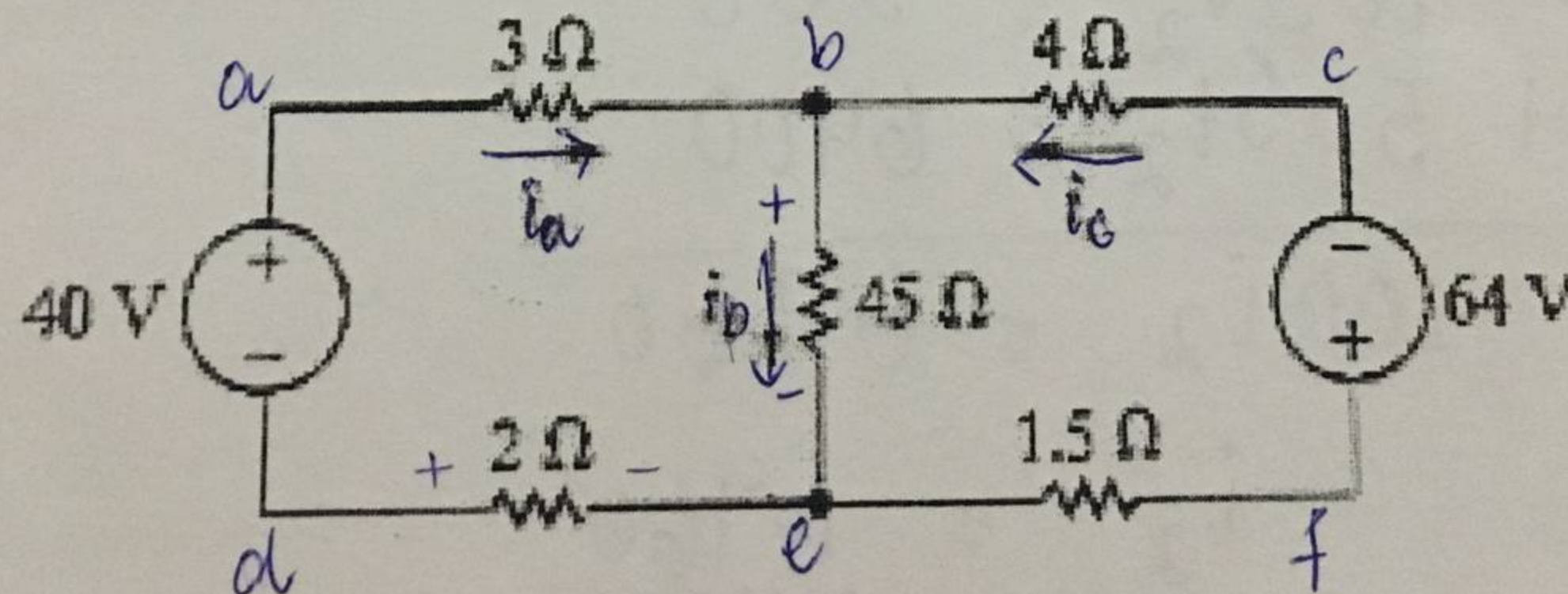
Marks: 10

OBL Statement:

The ability to apply node-voltage/mesh-current methods to find the unknowns in a circuit:

Problem

Use the mesh current method to find the branch currents i_a , i_b and i_c in the circuit:



Mesh abeda:

EE115 Linear Circuit Analysis (All sections)
Mid Term – II (FALL 2015)

Maximum Marks: 45

Instructor: Dr. Saima Zafar & Ms. Beenish Fatima

Total Time: 1 hr

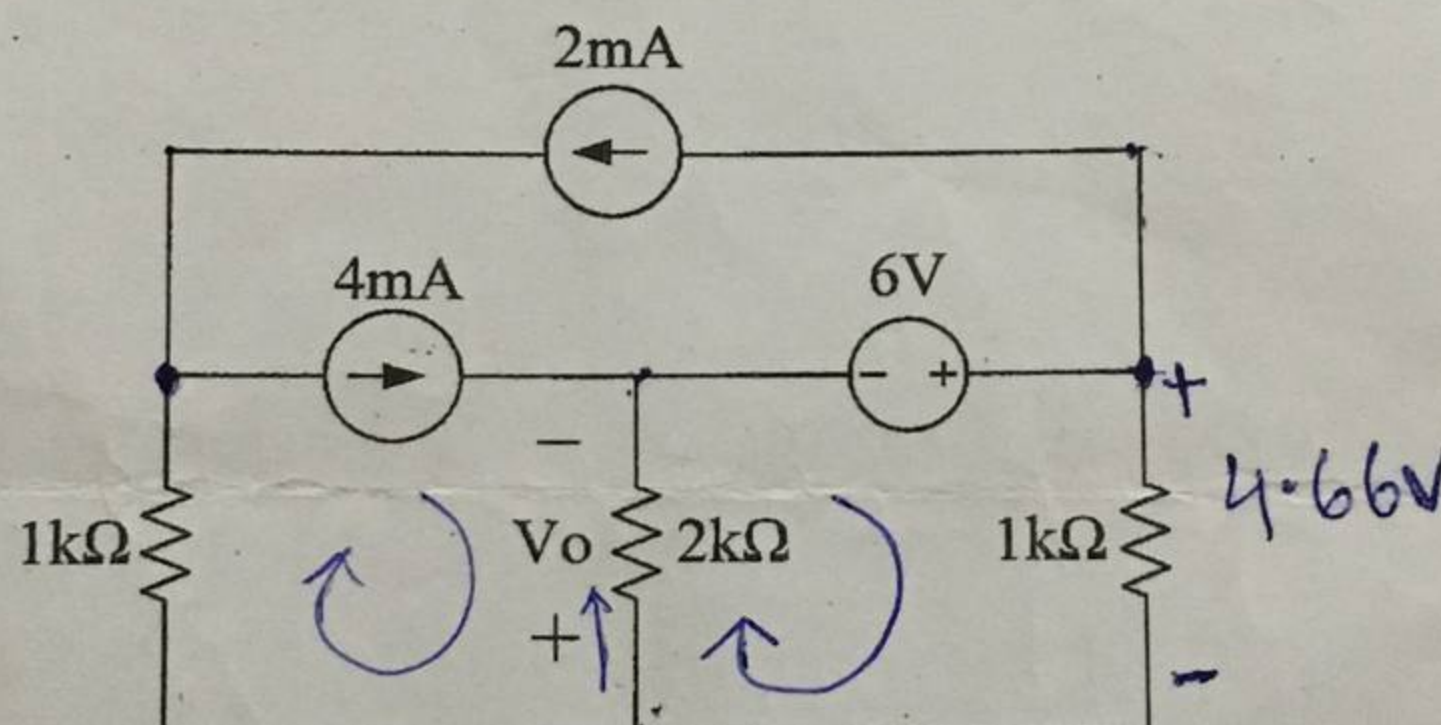
Dated: 7th November, 2015

INSTRUCTIONS: Carefully read the paper and attempt each question (only once) by using the technique mentioned in it. Show all calculations, direct answers are not acceptable. Make assumptions if you think some data is missing, but clearly mention your assumption. Your answers should be approximated up to two decimal places. Attempt all parts of a question together.

Question # 1 [15 marks]

Class Learning Objective: An ability to apply mesh current method to find the current/ voltage in a resistive circuit.

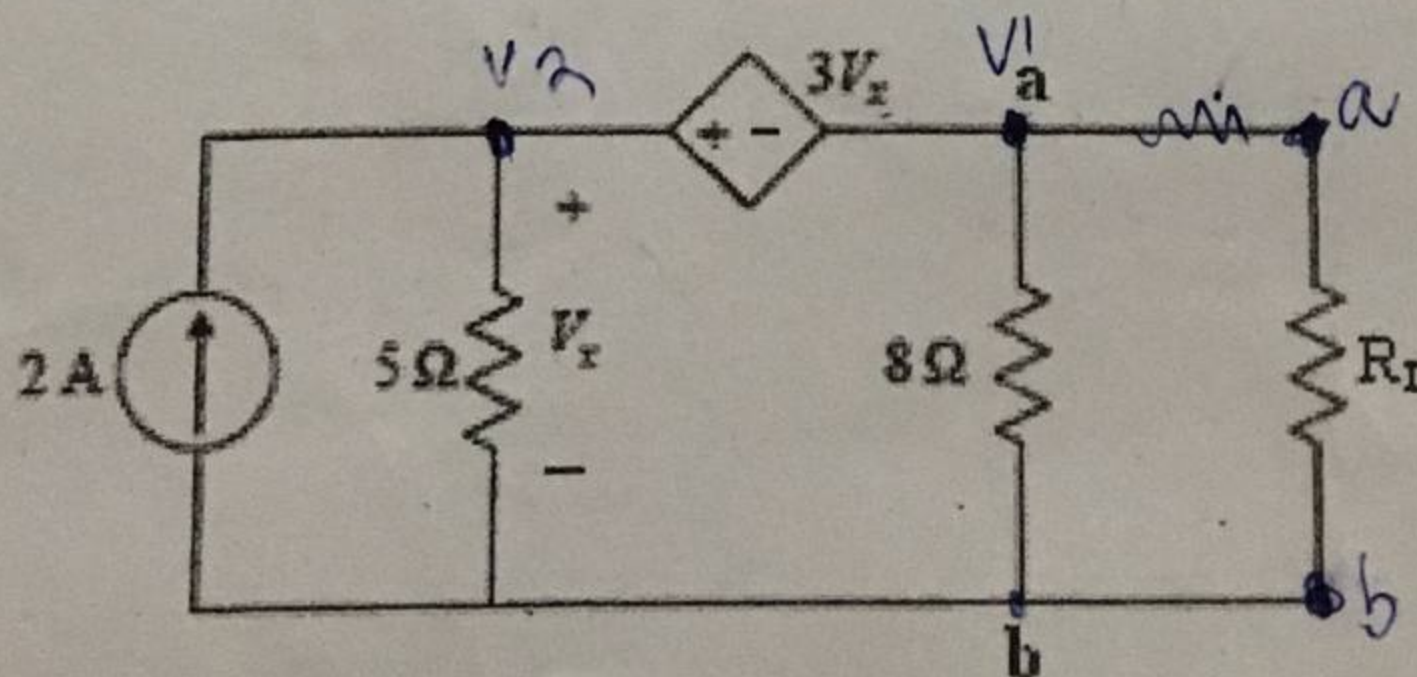
- Find V_0 in the circuit shown below using **mesh analysis**.
- Use the result of part (a) to find the power associated with the 4mA current source.



Question # 2 [15 marks]

Class Learning Objective: An ability to determine the Thevenin equivalent circuit and compute maximum power delivered to a load resistor.

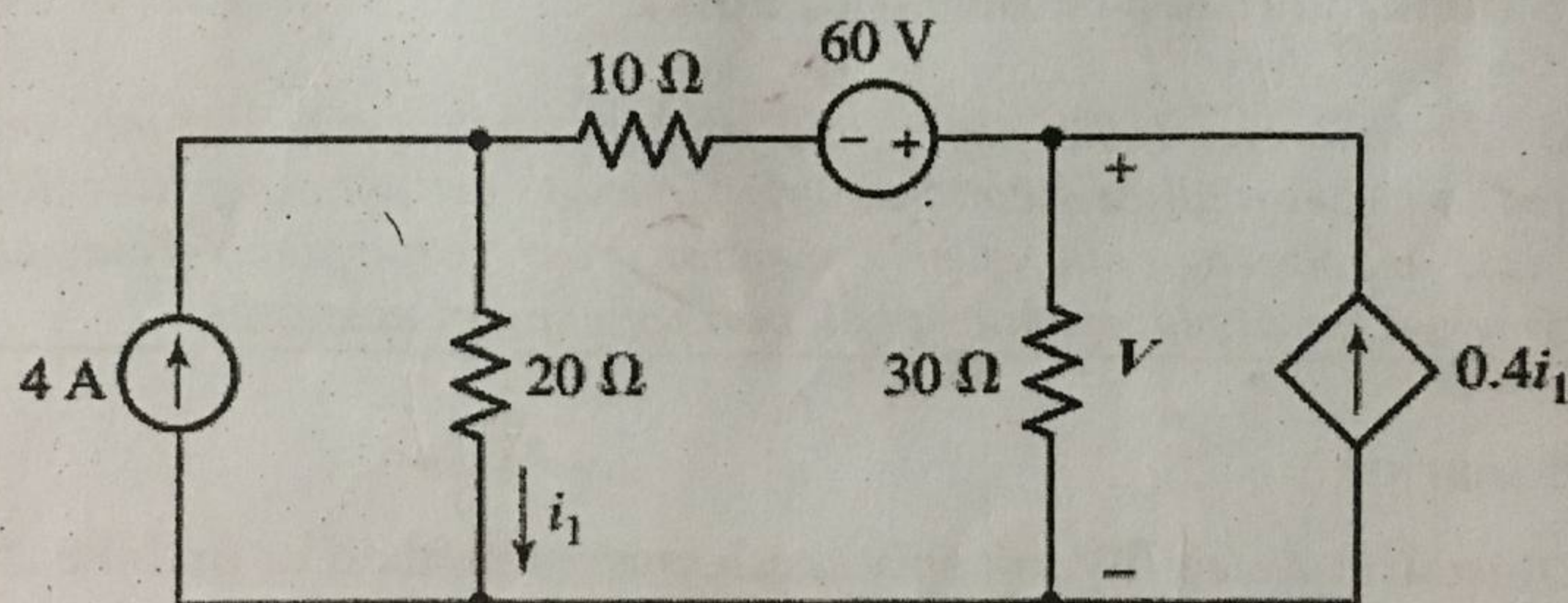
In the circuit shown below, find the value of R_L that will absorb maximum power from the circuit. Find also the value of this maximum power.



Question # 3 [15 marks]

Class Learning Objective: An ability to use the principle of superposition to find current/ voltage in a resistive circuit containing more than one independent source.

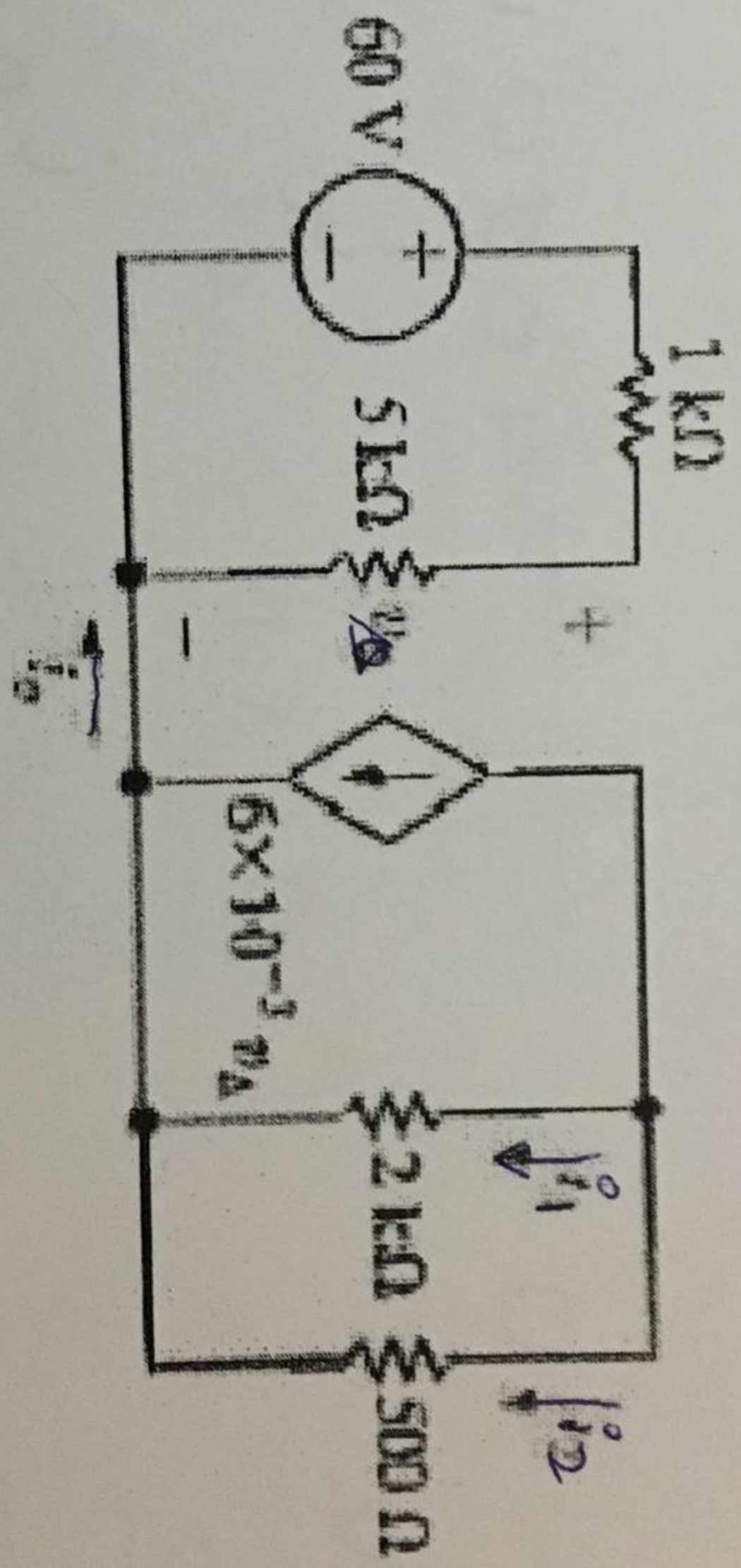
Use **superposition** to find the voltage V in the circuit given below:



	4		4	
				2

Problem

(a) Find the currents i_0 , i_1 and i_2 .



EE115 Linear Circuit Analysis (All sections)
Mid Term - I (FALL 2015)

Maximum Marks: 50

Instructor: Dr. Saima Zafar & Ms. Beenish Fatima

Total Time: 1 hr

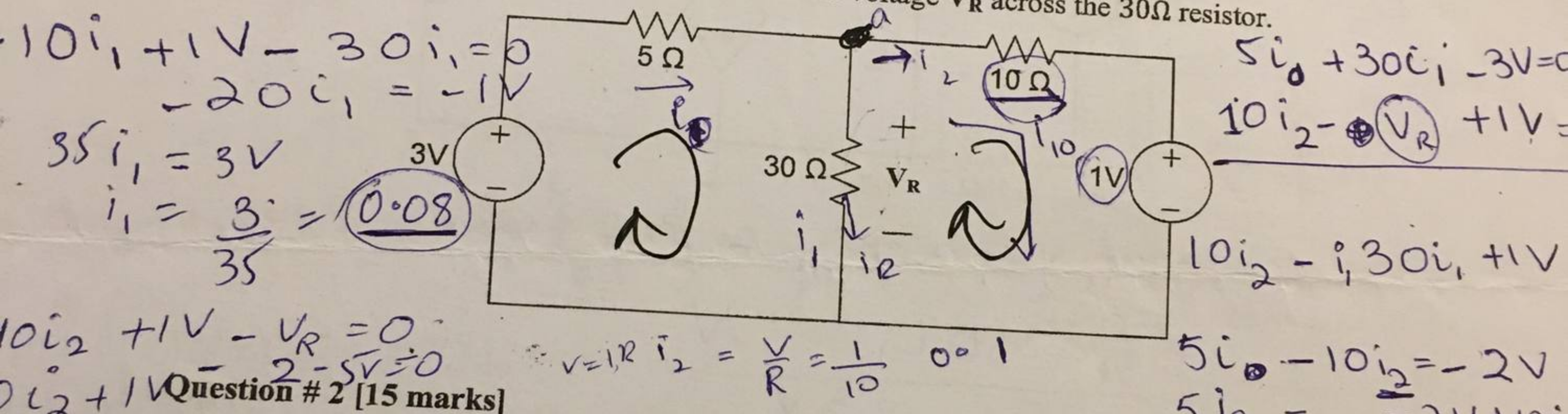
Dated: 15th September, 2015

INSTRUCTIONS: Carefully read the paper and attempt each question (only once) by using the technique mentioned in it. Show all calculations, direct answers are not acceptable. Make assumptions if you think some data is missing, but clearly mention your assumption. Your answers should be approximated up to two decimal places. Attempt all parts of a question together.

Question # 1 [15 marks]

Class Learning Objective: An ability to apply Ohm's Law and Kirchhoff's laws to analyze simple resistive circuits.

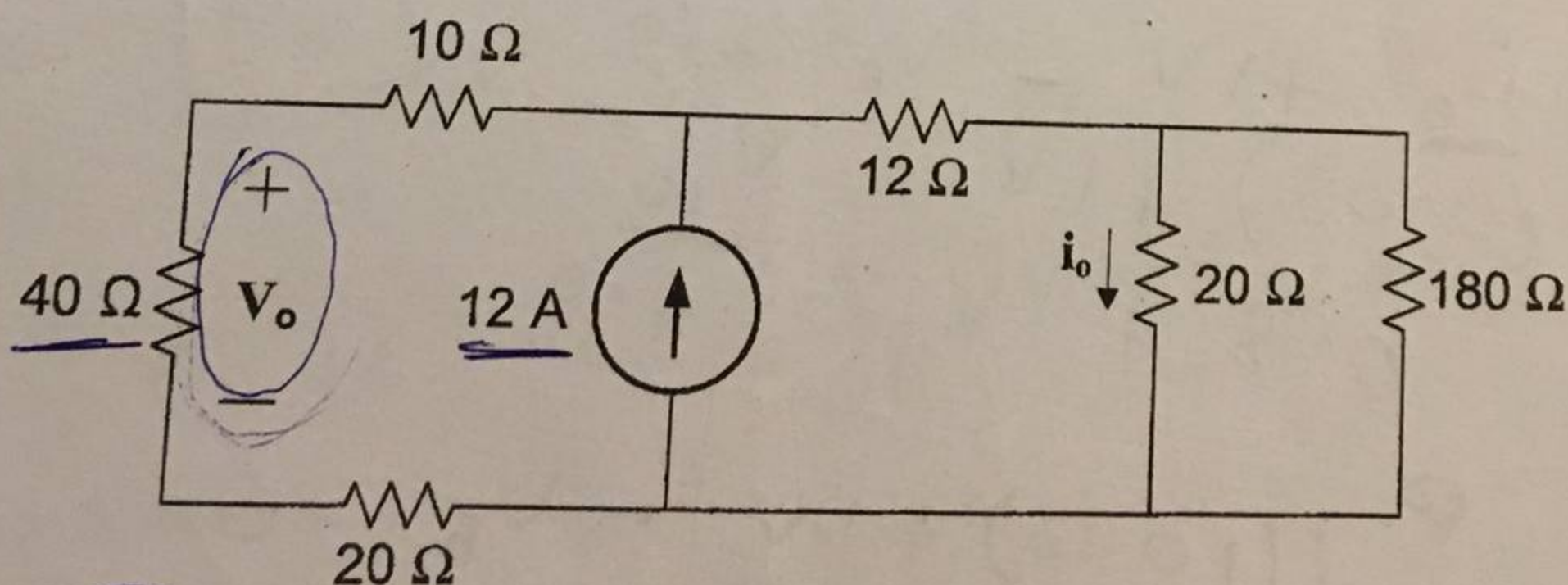
Use only **KCL, KVL** and **Ohm's Law** to find the voltage V_R across the 30Ω resistor.



Question # 2 [15 marks]

Class Learning Objective: An ability to simplify series, parallel circuits and apply CDR, VDR to compute unknown currents and voltages.

For the circuit shown in the figure below, use **current divider rule, voltage divider rule** and **Ohm's Law** only to calculate v_o , i_o and the **power** developed by the 12A current source.



$$R_i = \frac{V}{I} = \frac{V}{I/R}$$

$$\begin{aligned} 5i_1 + 30i_1 - 3V &= 0 \\ 10i_2 - V_R &+ 1V = 0 \\ -2V &= 0 \end{aligned}$$

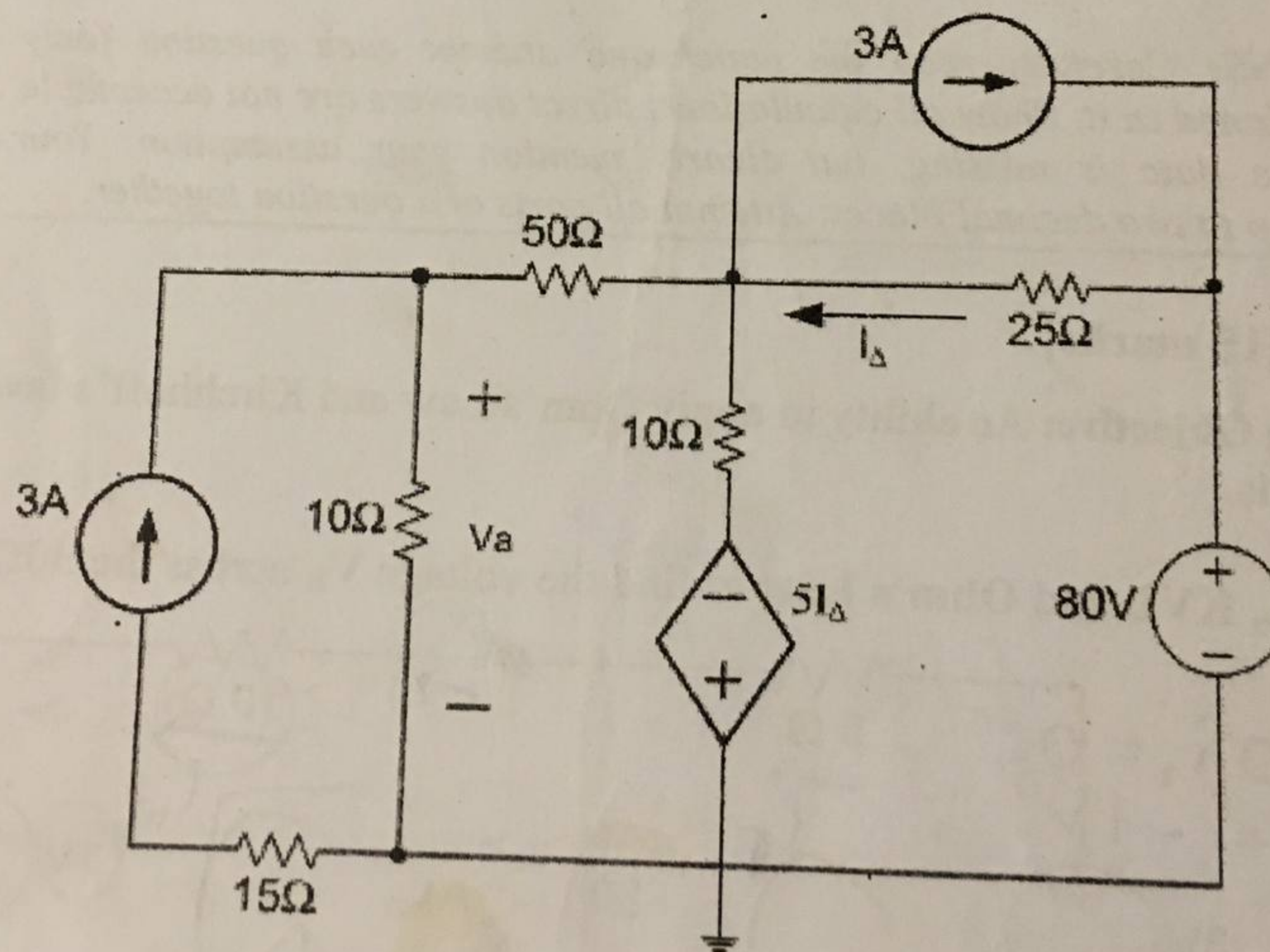
$$\begin{aligned} i_1 - i_2 + i_R + i_2 &= 0 \\ i_R &= i_1 - i_2 \\ &= 0.6 - 0.1 \end{aligned}$$

Question # 3 [16 + 4 = 20 marks]

Class Learning Objective: An ability to use node-voltage method to solve a circuit containing independent and/or dependent sources.

For the circuit given below, find the following using **Node Voltage** method:

- a) Voltage V_a
- b) Power associated with the 25Ω resistor.



$$(5\Omega)(i_\Delta) + 30i_\Delta - 3V = 0$$